LETTERS

Late formation and prolonged differentiation of the Moon inferred from W isotopes in lunar metals

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The Moon is thought to have formed from debris ejected by a giant impact with the early 'proto'-Earth[1] and, as a result of the high energies involved, the Moon would have melted to form a magma ocean. The timescales for formation and solidification of the Moon can be quantified by using ^{182}Hf–^{182}W and ^{146}Sm–^{142}Nd chronometry[2–4], but these methods have yielded contradicting results. In earlier studies[3,5–7], ^{182}W anomalies in lunar rocks were attributed to decay of ^{182}Hf within the lunar mantle and were used to infer that the Moon solidified within the first ~60 million years of the Solar System. However, the dominant ^{182}W component in most lunar rocks reflects cosmogenic production mainly by neutron capture of ^{181}Ta during cosmic-ray exposure of the lunar surface[3,7], compromising a reliable interpretation in terms of ^{182}Hf–^{182}W chronometry. Here we present tungsten isotope data for lunar metals that do not contain any measurable Ta-derived ^{182}W. All metals have identical ^{182}W/^{184}W ratios, indicating that the lunar magma ocean did not crystallize within the first ~60 Myr of the Solar System, which is no longer inconsistent with Sm–Nd chronometry[8–11]. Our new data reveal that the lunar and terrestrial mantles have identical ^{182}W/^{184}W. This, in conjunction with ^{147}Sm–^{143}Nd ages for the oldest lunar rocks[8–11], constrains the age of the Moon and Earth to 62^{+90}_{-10} Myr after formation of the Solar System. The identical ^{182}W/^{184}W ratios of the lunar and terrestrial mantles require either that the Moon is derived mainly from terrestrial material or that tungsten isotopes in the Moon and Earth's mantle equilibrated in the aftermath of the giant impact, as has been proposed to account for identical oxygen isotope compositions of the Earth and Moon[12].

We obtained tungsten isotope data for metals from two KREEP-rich samples (KREEP stands for enrichment in potassium (K), rare earth elements (REE) and phosphorus (P)), four low-Ti and five high-Ti mare basalts (Fig. 1 and Table 1). We processed fourfold to fivefold more material than an earlier study[3] and monitored the purity of our metal separates by determining their Hf/W ratios. These indicate that for the analyses reported here any possible contamination from silicate and oxide grains has no measurable effect on ^{182}W/^{184}W. Most of the samples investigated here had relatively short exposure times and required corrections of <0.1 ε units ($\varepsilon = 0.01\%$) for burnout of tungsten isotopes[13,14]; only for samples 15556 and 70017 (exposure ages ~220 and ~500 Myr) were corrections larger (~0.4 and ~0.2 ε units). Details of the corrections are given in Supplementary Information. All samples analysed here have identical ^{182}W/^{184}W ratios within ± 0.32 ε units (2 σ) and agree with previously reported data for metals from KREEP-rich samples[3]. Combined, these data average at ε^{182}W $= 0.09 \pm 0.10$ (2 s.e.m.), $n = 15$; ε^{182}W is defined in Table 1).

In contrast to earlier studies[3,5,6], we do not find ^{182}W/^{184}W variations within the lunar mantle. Lee *et al.*[5] reported ε^{182}W ≈ 1.4 for low-Ti mare basalt 15555; however, the exposure age of this sample, combined with its Sm isotopic composition and Ta/W ratio, indicates that this anomaly might be due entirely to cosmogenic ^{182}W. Kleine *et al.*[3] reported elevated ε^{182}W ≈ 2 for a magnetic separate from high-Ti mare basalt 79155 but we determined Hf/W = 7.5 for an aliquot from the same magnetic separate, most probably indicating the presence of some ilmenite and hence cosmogenic ^{182}W in this separate. The calculated cosmogenic ^{182}W component is ~1.7

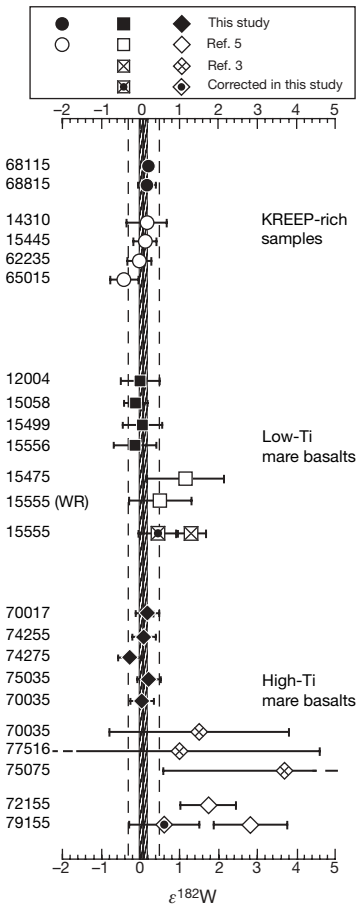

Figure 1 | ε^{182}W of lunar metals analysed in this study compared with data from refs 3 and 5. Some of the previous data (shown with black dots inside the symbols) were corrected for cosmogenic ^{182}W (see the text for details). Error bars indicate 2σ external reproducibilities. The hatched area indicates the average ε^{182}W $= 0.09 \pm 0.10$ (2 s.e.m., $n = 15$) of lunar metals from this study combined with previously reported data for metals from KREEP-rich samples[5]. The dashed lines indicate a 2σ of ± 0.32 ε^{182}W of these data.

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Table 1 | Hf–W data for lunar metals from KREEP-rich samples, low-Ti and high-Ti mare basalts

Sample	W (p.p.m.)	Hf (p.p.m.)	Hf/W	ε^{183}W	$\varepsilon^{182}W_{meas}$	$\varepsilon^{182}W_{corr}$
KREEP-rich samples						
68115	23.9	0.407	0.02	0.00 ± 0.10	0.33 ± 0.14	0.33 ± 0.14
				−0.01 ± 0.10	0.18 ± 0.12	0.18 ± 0.12
				0.01 ± 0.09	0.18 ± 0.12	0.18 ± 0.12
				0.00 ± 0.11	0.14 ± 0.15	0.14 ± 0.15
Mean (±2σ)				0.00 ± 0.02	0.21 ± 0.17	0.21 ± 0.17
68815	27.5	0.725	0.03	0.04 ± 0.11	0.02 ± 0.18	0.02 ± 0.18
				−0.02 ± 0.10	0.26 ± 0.18	0.26 ± 0.18
				0.03 ± 0.13	0.25 ± 0.15	0.25 ± 0.15
				−0.01 ± 0.13	0.22 ± 0.17	0.22 ± 0.17
Mean (± 2σ)				0.01 ± 0.06	0.18 ± 0.23	0.19 ± 0.23
Weighted average* (±2σ; n = 6)				0.00 ± 0.03	0.11 ± 0.21	0.11 ± 0.21
Low-Ti mare basalts						
12004	48.7	0.539	0.11	0.04 ± 0.21	0.00 ± 0.36	0.05 ± 0.36
15058	11.3	0.113	0.01	−0.10 ± 0.15	−0.11 ± 0.19	0.01 ± 0.20
15499	7.64	0.911	0.12	0.03 ± 0.22	0.06 ± 0.31	0.16 ± 0.31
15556	21.6	0.477	0.02	0.02 ± 0.21	−0.14 ± 0.29	0.30 ± 0.36
Weighted average (±2σ; n = 4)				−0.02 ± 0.09	−0.07 ± 0.13	0.09 ± 0.14
High-Ti mare basalts						
70017	11.6	2.18	0.19	−0.13 ± 0.14	0.20 ± 0.14	0.38 ± 0.16
70035	10.1	5.09	0.50	0.08 ± 0.13	0.05 ± 0.18	0.14 ± 0.18
74255	1.32	1.30	1.0	−0.11 ± 0.14	0.09 ± 0.17	0.11 ± 0.16
74275	8.92	2.24	0.25	−0.16 ± 0.18	−0.25 ± 0.25	−0.24 ± 0.25
75035	4.65	25.1	5.4	0.07 ± 0.15	0.22 ± 0.18	0.18 ± 0.18
Weighted average (±2σ; n = 5)				−0.04 ± 0.14	0.11 ± 0.19	0.16 ± 0.24
Bulk lunar mantle* (2 s.e.m., n = 15)				0.00 ± 0.02	0.02 ± 0.09	0.09 ± 0.10

$\varepsilon^{18i}W = 10^4 \times [(^{18i}W/^{184}W)_{sample}/(^{18i}W/^{184}W)_{standard} - 1]$. Replicates for KREEP-rich samples are repeated measurements of the same solution. Mean values are weighted averages calculated with Isoplot (n = number of samples). Errors are ±2σ unless indicated otherwise.
* Averages are calculated including data for KREEP-rich samples from ref. 3.

ε units; the elevated $^{182}W/^{184}W$ reported for 79155 is therefore presumably due mainly to cosmogenic ^{182}W. Enhanced $^{182}W/^{184}W$ in high-Ti mare basalts 72155 (ref. 3) and 75075 (ref. 5) are also not well resolved from $\varepsilon^{182}W = 0.09 \pm 0.10$ (2 s.e.m.). Details on the quantification of cosmogenic ^{182}W are given in the supplement. We conclude that in view of the identical $^{182}W/^{184}W$ we obtained for pure metal separates from nine mare basalts (average $\varepsilon^{182}W = 0.12 \pm 0.12$ (2 s.e.m.), $n = 9$), elevated $^{182}W/^{184}W$ ratios in mare basalts reported earlier probably reflect production of cosmogenic ^{182}W. This production is difficult to quantify, demonstrating that the purely radiogenic $^{182}W/^{184}W$ is difficult to determine precisely if cosmogenic ^{182}W is present. We conclude that there are no ^{182}Hf-induced tungsten isotope variations among KREEP and the mare basalt sources. Given that low-Ti and high-Ti mare basalts sample most cumulates from the lunar magma ocean (LMO)[15] and that KREEP contains a significant part of the incompatible element inventory of the lunar mantle[16], the average $\varepsilon^{182}W = 0.09 \pm 0.10$ (2 s.e.m.) of these lithologies most probably represents the average ε^{182}W of the lunar mantle. Although some ferroan anorthosites have elevated ε^{182}W values up to ~3 (ref. 6), these data are relatively imprecise and their weighted average $\varepsilon^{182}W = 1.9 \pm 1.7$ (2σ) is not resolvable from $\varepsilon^{182}W = 0.09 \pm 0.10$ (2 s.e.m.). Moreover, the analysed anorthosites might also contain cosmogenic ^{182}W.

These new constraints have far-reaching implications for the lifetime of the LMO as well as for the age and formation of the Moon. The homogeneous $^{182}W/^{184}W$ ratios of all lunar samples in spite of strongly fractionated Hf/W ratios in their source areas[17,18] indicate that the last equilibration of tungsten isotopes within the LMO occurred later than at ~60 Myr (here, Myr refers to time after formation of the first solids in the Solar System) (Fig. 2). Isotopic equilibration among the products of the LMO is possible up to a critical crystal fraction of ~60%, until which convection prevents crystal settling[19]. Although our new results only provide the earliest time (later than at ~60 Myr) for ~60% LMO crystallization, this new age is no longer in conflict with other constraints regarding the lifespan of the LMO. The rapid crystallization of the LMO required by the earlier tungsten isotope data[3,5,6] implied that some ^{147}Sm–^{143}Nd ages for ferroan anorthosites[8–11] and the ^{146}Sm–^{142}Nd model age[2,4] of the lunar mantle reflect post-LMO events[3,17]. With the revised Hf–W time constraint this is no longer required and the aforementioned Sm–Nd ages could possibly date processes associated with the LMO. This suggests that the LMO could have solidified to ~60% as late as at ~215 Myr, as given by its ^{146}Sm–^{142}Nd model age[2,4].

Our data constrain not only the lifespan of the LMO but also the timing of the giant impact. This uses the virtually identical

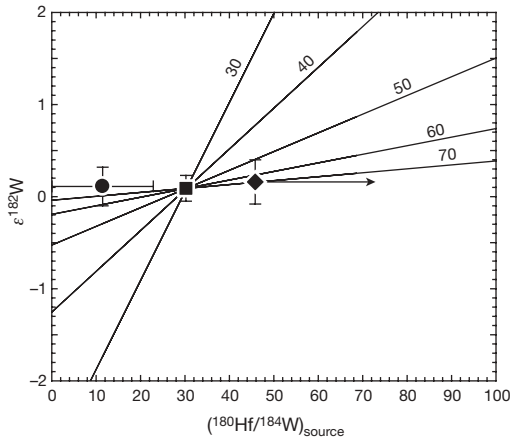

Figure 2 | Plot of ε^{182}W against source ^{180}Hf/^{184}W. Assumed Hf/W ratios are 10 ± 10 (2σ) for KREEP, 26.5 ± 1.1 (2σ) for the low-Ti mare basalt source, and >40 for the high-Ti mare basalt source. For details see Supplementary Information. ε^{182}W values shown for the low-Ti (square) and high-Ti (diamond) mare basalt sources are weighted averages of the data obtained in this study; the average ε^{182}W of KREEP (circle) was calculated by including data from ref. 3. Error bars indicate 2σ of these data. Reference isochrons corresponding to 30, 40, 50, 60 and 70 Myr, as indicated, after formation of the Solar System are shown. It is assumed that the Hf/W ratio of the bulk lunar mantle is similar to that of the low-Ti mare basalt source. The latter consists mainly of olivine and orthopyroxene, which are not capable of fractionating Hf and W (ref. 18). The identical ε^{182}W values for KREEP and the low-Ti and high-Ti mare basalt sources require that equilibration of tungsten isotopes within the LMO occurred later than ~60 Myr after the start of the Solar System. Note that this conclusion depends on neither the choice of Hf/W ratios of the sources nor on using the Hf/W ratio of the low-Ti mare basalt source as representative of the Hf/W ratio of the bulk lunar mantle.

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$^{182}W/^{184}W$ ratio of the bulk silicate Moon and Earth in conjunction with their different Hf/W ratios. The latter can be inferred from U/W and Th/W ratios because U, Th and W have similar incompatibilities during mantle melting[20,21]. The lunar mantle has U/W = 1.93 ± 0.08 (2σ)[21], distinctly higher than U/W = 1.3 ± 0.4 (2σ) for the bulk silicate Earth (obtained from Th/W = 5.5 ± 1.6 (2σ)[20] and Th/U = 4.2 ± 0.1 (2σ)[22]). If refractory lithophile elements occur in chondritic proportions in planetary mantles, then Hf/U = 13.7 for chondrites[23] can be used to calculate Hf/W = 26.5 ± 1.1 (2σ) and Hf/W = 18.0 ± 5.2 (2σ) for the bulk silicate Moon and Earth. However, it is conceivable that the Moon and Earth have non-chondritic ratios of refractory lithophile elements (for example, Th/U (refs 22, 23)) and in this case the Hf/W ratios of the bulk silicate Moon and Earth might be different from those calculated above. However, this would have only a small effect on the calculated ages (Fig. 3). For instance, using a ~15% higher Hf/U ratio changes the ages by only ~2 Myr.

A first-order estimate for core formation in the Moon (or the impactor) is obtained using a two-stage model. This assumes a chondritic initial $^{182}W/^{184}W$ of the Moon and results in an age of ~37 Myr. This model implies that the bulk silicate Earth and Moon started off with slightly different $^{182}W/^{184}W$ ratios (~0.5 ε units difference at most; see Supplementary Fig. 1) and fortuitously evolved to identical present-day $^{182}W/^{184}W$ ratios. However, as the Moon consists predominantly of mantle material with high Hf/W and hence most probably radiogenic $^{182}W/^{184}W$, this two-stage model is not valid. The initial $^{182}W/^{184}W$ of the Moon was most probably higher than chondritic, resulting in an age younger than ~37 Myr and implying that the bulk silicate Moon and Earth must have had indistinguishable initial $^{182}W/^{184}W$ ratios (Supplementary Fig. 1).

A more reliable age constraint for core formation is obtained from the identical $^{182}W/^{184}W$ ratios of the bulk silicate Moon and Earth in conjunction with their distinct Hf/W ratios and by assuming identical initial $^{182}W/^{184}W$ ratios for the bulk silicate Moon and Earth. As shown in Fig. 3, core formation in the Moon must have occurred later than at ~50 Myr, otherwise the lunar mantle would have a ^{182}W excess relative to the terrestrial mantle. This new constraint provides the earliest time that the giant impact could have occurred because the segregation of the lunar core probably occurred briefly after formation of the Moon. This also holds true if the Hf–W systematics of the Moon would largely reflect core formation in the impactor because this must pre-date the giant impact.

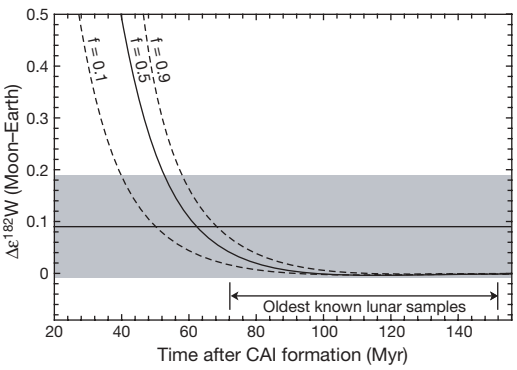

Figure 3 | $\varepsilon^{182}W$ difference between bulk silicate Moon and Earth as a function of time of core formation. The grey shaded area indicates $\varepsilon^{182}W = 0.09 \pm 0.10$ (2 s.e.m.) for the lunar mantle as determined in this study. Assuming that the lunar and terrestrial mantles had identical initial $^{182}W/^{184}W$ ratios, the time at which these two reservoirs separated can be calculated from

$$t = \frac{1}{\lambda} \times \ln\left\{\frac{\frac{\Delta\varepsilon^{182}W \times 10^{-4}}{(^{182}W/^{184}W)_{BSE}} \Big/ \left\{1.14 \times (Hf/U)_{CHUR} \times \left[(U/W)_{BSE} - (U/W)_{BSM}\right]\right\}}{(^{182}Hf/^{180}Hf)_0}\right\}$$

where $(^{182}Hf/^{180}Hf)_0 = 1.07 \times 10^{-4}$ (ref. 14). CHUR, chondritic uniform reservoir; BSE, bulk silicate Earth; BSM, bulk silicate Moon.

$f_{Hf/W} = \left\{(Hf/W)_{BSM}/(Hf/W)_{BSE} - 1\right\}$. Note that the lower uncertainty on the age is calculated by propagating the uncertainties on $\Delta\varepsilon^{182}W$ and $^{180}Hf/^{184}W$ and therefore does not coincide with the intersection of the $f = 0.1$ curve and the grey shaded bar. The double-headed arrow indicates the ages of the oldest known lunar samples (4.46 ± 0.04 Gyr). CAI, calcium aluminium-rich inclusion.

Because of the identical $^{182}W/^{184}W$ ratios of the bulk silicate Moon and Earth, our data provide only the earliest time at which the Moon could have formed, but the latest time is given by the age of the oldest lunar rocks. Combined with the 4.456 ± 0.040 Gyr (2σ) ^{147}Sm–^{143}Nd age of the lunar crust[10] we obtain an age of 62^{+90}_{-10} Myr for the formation of the Moon (Fig. 3). Given that Earth's accretion cannot have terminated before the giant impact, this also provides an age for the accretion of Earth. This is inconsistent with termination of Earth's accretion at ~30 Myr (for example, ref. 24) and is also difficult to reconcile with the <30 Myr ^{146}Sm–^{142}Nd model age for the differentiation of Earth's mantle[25]. Our new age constraint, however, is consistent with most U–Pb model ages for the Earth[26].

The identical $^{182}W/^{184}W$ ratios of the lunar and terrestrial mantles provide a key constraint on the formation of the Moon. The $^{182}W/^{184}W$ of any planetary mantle reflects the timescale of accretion and core formation, the degree of tungsten depletion, and the extent of re-equilibration of tungsten isotopes during core formation[24,27,28]. It is therefore unlikely that the mantles of the proto-Earth and the impactor evolved to identical $^{182}W/^{184}W$ ratios. Successful simulations of the giant impact predict that ~80% of the Moon is derived from impactor material[1], such that small pre-existing tungsten isotopic differences between proto-Earth and impactor should be reflected in the composition of the Moon. However, this is not observed in the Hf–W systematics. Similarly, the identical oxygen isotopic compositions of the Earth and Moon in spite of widespread oxygen isotopic heterogeneity among objects of the inner Solar System are unexpected. These have been interpreted to reflect accretion of the Earth and Moon from a similar mix of components formed at the same heliocentric distance[29]. As explained above, such a model cannot account for the identical tungsten isotope compositions of the lunar and terrestrial mantles. Hence, either the Moon is derived almost entirely from Earth's mantle (which is contrary to results from numerical simulations of the giant impact) or lunar and terrestrial materials equilibrated in the aftermath of the giant impact. Diffusive exchange between the silicate vapour atmosphere of the proto-Earth and the lunar magma disk might be possible for elements that became vaporized during formation of the lunar magma disk. It has been shown that this is possible for oxygen isotopes[12], but the efficiency to which tungsten became vaporized and hence could have been equilibrated isotopically remains to be investigated.

METHODS SUMMARY

Samples were crushed in an agate mortar and magnetic fractions were obtained with a hand magnet. The magnetic separates were purified by repeated grinding, magnetic separation and ultrasonication in distilled ethanol. After dissolution a ~5% aliquot was spiked with a mixed ^{180}Hf–^{183}W tracer for the determination of hafnium and tungsten concentrations that are used to monitor the purity of the metal separates. From the remaining ~95%, tungsten was extracted with the use of anion exchange techniques, slightly modified from refs 3, 30. All measurements were performed with a Nu Plasma MC-ICPMS at ETH Zürich. Tungsten isotope measurements were normalized to $^{186}W/^{183}W = 1.9859$ and $^{186}W/^{184}W = 0.92767$ by using the exponential law. Results obtained with these two normalization procedures agree for all samples. Tungsten isotope compositions of the samples were determined relative to the $^{182}W/^{184}W$ ratios obtained for two bracketing measurements of the ALFA AESEAR tungsten standard solution. Isobaric Os interferences on masses 186 and 184 were monitored by measuring ^{188}Os but corrections were insignificant for all samples (<0.01 p.p.m.). The accuracy of the measurements was monitored by using the $^{183}W/^{184}W$ ratio, and all samples have $^{183}W/^{184}W$ ratios identical to the standard value to within

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$\pm 0.2\ \varepsilon\ (2\sigma)$ units. The external reproducibility of the tungsten isotope measurements was evaluated by repeated measurements of metals from the two KREEP-rich samples and was ± 0.17 (2σ, $n = 4$) and ± 0.23 (2σ, $n = 4$) (Table 1). For the low-Ti mare basalts, the reproducibility was $\sim 0.5\ \varepsilon$ units or better, and for the high-Ti mare basalts it is 0.3–0.4 ε units.

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1. Canup, R. M. & Asphaug, E. Origin of the Moon in a giant impact near the end of the Earth's formation. *Nature* **412,** 708–712 (2001).
2. Nyquist, L. E. *et al.* Sm-146–Nd-142 formation interval for the lunar mantle. *Geochim. Cosmochim. Acta* **59,** 2817–2837 (1995).
3. Kleine, T., Palme, H., Mezger, K. & Halliday, A. N. Hf–W chronometry of lunar metals and the age and early differentiation of the Moon. *Science* **310,** 1671–1674 (2005).
4. Rankenburg, K., Brandon, A. D. & Neal, C. R. Neodymium isotope evidence for a chondritic composition of the Moon. *Science* **312,** 1369–1372 (2006).
5. Lee, D. C., Halliday, A. N., Leya, I., Wieler, R. & Wiechert, U. Cosmogenic tungsten and the origin and earliest differentiation of the Moon. *Earth Planet. Sci. Lett.* **198,** 267–274 (2002).
6. Lee, D. C., Halliday, A. N., Snyder, G. A. & Taylor, L. A. Age and origin of the moon. *Science* **278,** 1098–1103 (1997).
7. Leya, I., Wieler, R. & Halliday, A. N. Cosmic-ray production of tungsten isotopes in lunar samples and meteorites and its implications for Hf–W cosmochemistry. *Earth Planet. Sci. Lett.* **175,** 1–12 (2000).
8. Borg, L. E. *et al.* Isotopic studies of ferroan anorthosite 62236: A young lunar crustal rock from a light rare-earth-element-depleted source. *Geochim. Cosmochim. Acta* **63,** 2679–2691 (1999).
9. Carlson, R. W. & Lugmair, G. W. The age of ferroan anorthosite 60025: oldest crust on a young Moon? *Earth Planet. Sci. Lett.* **90,** 119–130 (1988).
10. Norman, M. D., Borg, L. E., Nyquist, L. E. & Bogard, D. D. Chronology, geochemistry, and petrology of a ferroan noritic anorthosite clast from Descartes breccia 67215: Clues to the age, origin, structure, and impact history of the lunar crust. *Meteorit. Planet. Sci.* **38,** 645–661 (2003).
11. Nyquist, L. *et al.* Feldspathic clasts in Yamato-86032: Remnants of the lunar crust with implications for its formation and impact history. *Geochim. Cosmochim. Acta* **70,** 5990–6015 (2006).
12. Pahlevan, K. & Stevenson, D. J. Equilibration in the aftermath of the lunar-forming giant impact. *Earth Planet. Sci. Lett.* **262,** 438–449 (2007).
13. Leya, I., Wieler, R. & Halliday, A. N. The influence of cosmic-ray production on extinct nuclide systems. *Geochim. Cosmochim. Acta* **67,** 529–541 (2003).
14. Kleine, T., Mezger, K., Palme, H., Scherer, E. & Münker, C. Early core formation in asteroids and late accretion of chondrite parent bodies: Evidence from ^{182}Hf–^{182}W in CAIs, metal-rich chondrites and iron meteorites. *Geochim. Cosmochim. Acta* **69,** 5805–5818 (2005).
15. Shearer, C. K. & Papike, J. J. Magmatic evolution of the Moon. *Am. Mineral.* **84,** 1469–1494 (1999).
16. Palme, H. & Wänke, H. A unified trace-element model for the evolution of the lunar crust and mantle. *Proc. Lunar Sci. Conf. 6th* 1179–1202 (1975).
17. Shearer, C. K. & Newsom, H. E. W–Hf isotope abundances and the early origin and evolution of the Earth–Moon system. *Geochim. Cosmochim. Acta* **64,** 3599–3613 (2000).
18. Righter, K. & Shearer, C. K. Magmatic fractionation of Hf and W: Constraints on the timing of core formation and differentiation in the Moon and Mars. *Geochim. Cosmochim. Acta* **67,** 2497–2507 (2003).
19. Solomatov, V. S. in *Origin of the Earth and Moon* (eds Canup, R. M. & Righter, K.) 323–338 (Lunar and Planetary Institute, Houston, TX, 2000).
20. Newsom, H. E. *et al.* The depletion of W in the bulk silicate Earth: constraints on core formation. *Geochim. Cosmochim. Acta* **60,** 1155–1169 (1996).
21. Palme, H. & Rammensee, W. The significance of W in planetary differentiation processes: Evidence from new data on eucrites. *Proc. 12th Lunar Planet. Sci. Conf.* 949–964 (1981).
22. Allègre, C. J., Dupre, B. & Lewin, E. Thorium uranium ratio of the Earth. *Chem. Geol.* **56,** 219–227 (1986).
23. Rocholl, A. & Jochum, K. P. Th, U and other trace elements in carbonaceous chondrites: Implications for the terrestrial and solar-system Th/U ratios. *Earth Planet. Sci. Lett.* **117,** 265–278 (1993).
24. Jacobsen, S. B. The Hf–W isotopic system and the origin of the Earth and Moon. *Annu. Rev. Earth Planet. Sci.* **33,** 531–570 (2005).
25. Boyet, M. & Carlson, R. W. ^{142}Nd evidence for early (>4.53 Ga) global differentiation of the silicate Earth. *Science* **309,** 576–581 (2005).
26. Allègre, C. J., Manhes, G. & Gopel, C. The age of the Earth. *Geochim. Cosmochim. Acta* **59,** 1445–1456 (1995).
27. Kleine, T., Mezger, K., Palme, H., Scherer, E. & Münker, C. The W isotope evolution of the bulk silicate Earth: constraints on the timing and mechanisms of core formation and accretion. *Earth Planet. Sci. Lett.* **228,** 109–123 (2004).
28. Nimmo, F. & Agnor, C. B. Isotopic outcomes of *N*-body accretion simulations: Constraints on equilibration processes during large impacts from Hf/W observations. *Earth Planet. Sci. Lett.* **243,** 26–43 (2006).
29. Wiechert, U. *et al.* Oxygen isotopes and the moon-forming giant impact. *Science* **294,** 345–348 (2001).
30. Kleine, T., Mezger, K., Palme, H., Scherer, E. & Münker, C. The W isotope composition of eucrites metal: Constraints on the timing and cause of the thermal metamorphism of basaltic eucrites. *Earth Planet. Sci. Lett.* **231,** 41–52 (2005).

Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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